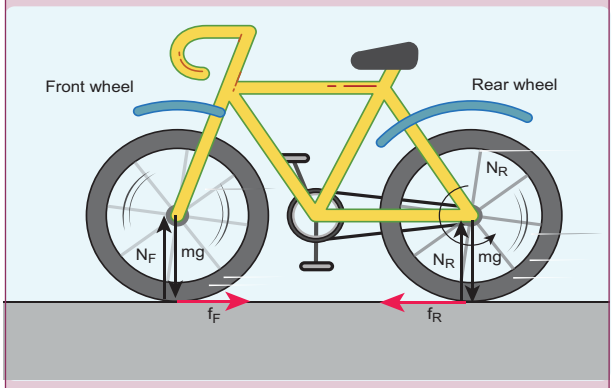
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Frictional force in the motion of a bicycle

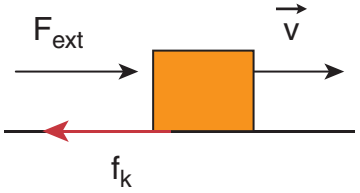
When a bicycle moves in the forward direction, what is the direction of frictional force in the rear and front wheels?



When we pedal a bicycle, we try to push the surface backward and the velocity of point of contact in the rear wheel is backwards. So, the frictional force pushes the rear wheel to move forward. But as the front wheel is connected with a rigid support to the back wheel, the forward motion of back wheel pushes the front wheel in the forward direction. So, the frictional forces act backward. Remember both frictional forces correspond to only static friction and not kinetic friction. If the wheel slips then kinetic friction comes into effect. In addition to static friction, the rolling friction also acts on both wheels in the backward direction.

EXAMPLE 3.21

Consider an object moving on a horizontal surface with a constant velocity. Some external force is applied on the object to keep the object moving with a constant velocity. What is the net force acting on the object?



Solution

If an object moves with constant velocity, then it has no acceleration. According to Newton's second law there is no net force acting on the object. The external force is balanced by the kinetic friction.

Note

It is not that 'no force acts on the object'. In fact there are two forces acting on the object. Only the net force acting on the object is zero.

3.7 DYNAMICS OF CIRCULAR MOTION

In the previous sections we have studied how to analyse linear motion using Newton's laws. It is also important to know how to apply Newton's laws to circular motion, since circular motion is one of the very common types of motion that we come across in our daily life. A particle can be in linear motion with or without any external force. But when circular motion occurs there must necessarily be some force acting on the object. There is no Newton's first law for circular motion. In other words without a force, circular motion cannot occur in nature. A force can change the velocity of a particle in three different ways.

1. The magnitude of the velocity can be changed without changing the direction of the velocity. In this case the particle will move in the same direction but with acceleration.